

Activity - 05

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1 Qn) given input {4371, 1323, 6173, ⁴¹⁹⁹4344, 9679, 1989}
 and a hash function $h(x) = x \pmod{10}$
 show the resulting

- separate chaining hash table
- open addressing hash table using quadratic Linear
- open addressing hash table using quadratic
- open addressing hash table with second hash function $h_2(x) = 7 - (x \pmod{7})$

Sol: a) separate chaining:-

$$h(x) = x \pmod{10}$$

$$h(4371) = 4371 \% 10 = 1$$

$$h(1323) = 1323 \% 10 = 3$$

$$h(6173) = 6173 \% 10 = 3$$

$$h(4199) = 4199 \% 10 = 9$$

$$h(4344) = 4344 \% 10 = 4$$

$$h(9679) = 9679 \% 10 = 9$$

$$h(1989) = 1989 \% 10 = 9$$

index

0	
1	→ 4371
2	
3	→ 6173 → 1323
4	→ 4344
5	
6	
7	
8	
9	→ 4199 → 9679 → 1989

b) Linear Hashing:-

$$h(x) = h(\text{key} + 1)$$

$$1) h(4371) = 4371 \% 10 = 1$$

$$2) h(1323) = 1323 \% 10 = 3$$

$$3) h(6173) = 6173 \% 10 = 3 \Rightarrow \text{collision}$$

$$h(6173) = h(6173 + 10) = h(6173) \% 10 = 3$$

collision. a 6 p 21

$$h(6173) = h(6173 + 1) = h(6174) \Rightarrow 10 = 4 \Rightarrow P = 2$$

$$4) h(4199) = h(4199) \cdot 10 = 9 //$$

$$5) h(4344) = h(4344) \cdot 10 = 4 \Rightarrow \text{collision}$$

$$h(4344) = h(4344 + 0) = h(4344) \cdot 10$$

$$= 4 \Rightarrow P = 1 \text{ collision}$$

$$h(4344) = h(4344 + 1) = h(4345) \cdot 10 = 5$$

$$6) h(9679) = h(9679) \cdot 10 = 9$$

$$\text{collision}$$

$$h(9679) = h(9679 + 1) \cdot 10 = h(9680) \cdot 10 = 0 \Rightarrow P = 2$$

$$7) h(1989) = h(1989) \cdot 10 = 9$$

$$\text{collision}$$

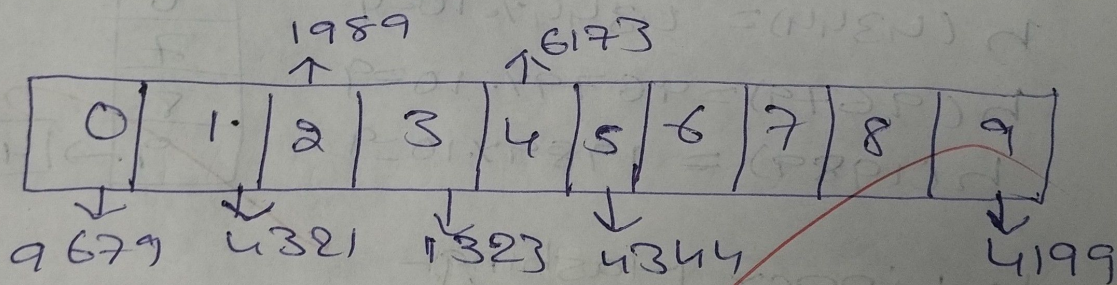
$$= h(1989 + 1) \cdot 10 = h(1990) \cdot 10 = 0 \Rightarrow P = 2$$

$$\text{collision}$$

$$= h(1989 + 2) \cdot 10 = h(1991) \cdot 10 = 1$$

$$\text{collision}$$

$$= h(1989 + 3) \cdot 10 = h(1992) \cdot 10 = 2 //$$



c) quadratic probing

$$h(\text{key}) = h(\text{key} + i^2) \Rightarrow i = 1, 2, 3 \dots$$

$$1) h(4371) = 4371 \cdot 1 \cdot 10 = 1$$

$$2) h(1323) = 1323 \cdot 1 \cdot 10 = 3$$

$$3) h(6173) = 6173 \cdot 1 \cdot 10 = 3$$

$$h(6173) = h(6173) \cdot 1 \cdot 10 = 3 \quad \text{collision}$$

$$h(6174) \cdot 1 \cdot 10 = 4$$

$$4) h(4199) = 4199 \cdot 1 \cdot 10 = 9$$

$$5) h(4344) = 4344 \cdot 1 \cdot 10 = 4$$

collision.

$$h(4344) = h(4344 + 1^2) \cdot 1 \cdot 10 = 5$$

$$6) h(9679) = 9679 \cdot 1 \cdot 10 = 9$$

collision.

$$h(9679) = h(9679 \cdot 1 \cdot 10 = 9$$

collision

$$h(9679) = h(9679 + 1^2) \cdot 1 \cdot 10$$

$$= h(9780) \cdot 1 \cdot 10 = 0$$

$$7) h(1989) = h(1989) \cdot 1 \cdot 10 = 9$$

collision

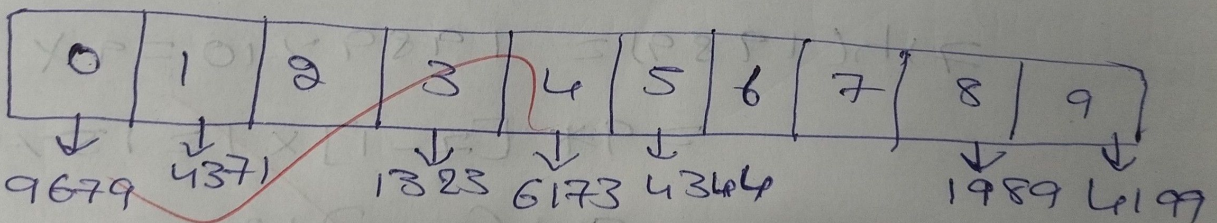
$$= h(1989 + 1) = h(1990) \cdot 1 \cdot 10 = 0$$

collision.

$$h(1989 + 2^2) = h(1993) \cdot 1 \cdot 10 = 3$$

collision

$$h(1989 + 3^2) = h(1998) \cdot 1 \cdot 10 = 8$$



d) Double hashing :-

$$h(\text{key}) = h_1(\text{key}) + [h_2(\text{key}) * i] \text{ mod size}$$

$$1) h(4371) = 4371 \div 10 = 1$$

$$2) h(1323) = 1323 \div 10 = 3$$

$$3) h(6173) = 6173 \div 10 = 3 \text{ collision}$$

$$h(6173) = 3 + (7-6) \times 1$$
$$= 3 + 1 = 4$$

$$4) h(4199) = 4199 \div 10 = 9$$

$$5) h(4344) = 4344 \div 10 = 4 \times$$

$$h(4344) = 4 + (7-4) \times 1$$

$$= 4 + 3 = 7$$

$$6) h(9679) = 9679 \div 10 = 9 \times$$

$$= 9 + (7-5) \times 1$$

$$= 11 \div 10 = 1 \text{ collision}$$

$$= 9 + (7-5) \times 2$$

$$\Rightarrow 13 \div 10 = 3 \text{ collision}$$

$$= 9 + (7-5) \times 3$$

$$= 15 \div 10 = 5 \times$$

$$7) h(1989) = 1989 \div 10 = 9 \times$$

$$= 9 * [7-1] \times 1 = 15 \div 10 = 5 \times$$

$$= 9 + (7-1) \times 2$$

$$= 21 \div 10 = 1 \times$$

$$= 9 + (7-1) \times 3$$

$$= 29 \div 10 = 2 \times$$

we cant Insert

0	1	2	3	4	5	6	7	8	9
↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
4371	1323	6173				4344	4199		

cant insert 1989 Doesn't
have the Index position Because
(i) value extended the ^{table} value

Good
Amir
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